Predictive Modelling for Building Energy Management

IE 7275 - Data Mining in Engineering

Northeastern University
Maana Javadi

June 2026

Table of Contents

Problem Setting

Buildings are among the largest contributors to urban greenhouse gas emissions, accounting for over one-third of emissions in many major cities (United Nations Environment Programme, 2025). As governments pursue climate targets, understanding what makes a building efficient and being able to identify inefficient buildings has become an essential tool for policy and urban planning.

This challenge is central to Canada's growing network of building energy benchmarking and disclosure programs. At the federal level, Natural Resources Canada (NRCan) administers ENERGY STAR Portfolio Manager, the primary tool used across the country to measure and benchmark building energy performance (Natural Resources Canada, 2025). These programs aim to create market demand for energy efficiency by making building performance transparent and supporting cities in tracking progress toward emissions-reduction goals. However, with thousands of buildings reporting each year, identifying the structural and operational factors that distinguish high-performing buildings from low-performing ones remains a significant analytical challenge.

Problem Definition

This project addresses two research questions:

1. How accurately can a building's ENERGY STAR score be predicted from its characteristics, and which features most strongly drive that score?
2. Can buildings be reliably classified as passing or failing the ENERGY STAR certification threshold (a score of 75 or above) from those same characteristics, and which features drive that classification?

These questions can help inform building energy policy by identifying which factors most strongly drive energy performance. Feature importance outputs from the classification model can be used to highlight the characteristics most associated with passing the

ENERGY STAR threshold, helping building owners and policymakers take a more targeted approach to retrofits and efficiency improvements.

Data Sources

The dataset used in this project will be the 2015 Building Energy Benchmarking data published by the City of Seattle on Kaggle (City of Seattle, 2016).

Although this dataset originates from a U.S. city, the methodology developed here is directly transferable to Canadian buildings. Both countries use the same benchmarking framework (ENERGY STAR Portfolio Manager, developed by the U.S. Environmental Protection Agency (EPA) and adapted for Canadian use by Natural Resources Canada) meaning the ENERGY STAR score, site EUI, and reported building characteristics are measured consistently across both jurisdictions.

Data Description

The raw dataset contains 3,340 rows (buildings) and 47 columns. Columns span four broad groups: identifiers and administrative codes; physical/structural attributes; energy-use measurements; and the target. The regression target is ENERGYSTARScore and the classification target is a derived binary column, ENERGYSTARCertification (Pass if score >= 75, else Fail).

A sample of variable names by category:

Category	**Example variables**
Identifiers / admin	OSEBuildingID, PropertyName, ComplianceStatus, CouncilDistrictCode, ZipCodes
Physical attributes	BuildingType, PrimaryPropertyType, Neighborhood, YearBuilt, NumberofFloors,

	PropertyGFATotal, PropertyGFAParking, Location (lat/long)
Energy use	SiteEUI(kBtu/sf), SourceEUI(kBtu/sf), SiteEnergyUse(kBtu), Electricity(kBtu), NaturalGas(kBtu), GHGEmissions(MetricTonsCO2e)
Target	ENERGYSTARScore (1-100); derived ENERGYSTARCertification (Pass/Fail)

Data Exploration

Statistical Methods

A first glance at the dataset (using df.head()) identified the ID columns and other columns that would not be useful for modelling (e.g., “DataYear” which was a constant 2015). Following this, a dataframe listing all the columns with their data types and percent of missing values was used to 1) visualize all columns, 2) identify any potential columns of suspicious data types, and 3) identify columns with a large number of missing values.

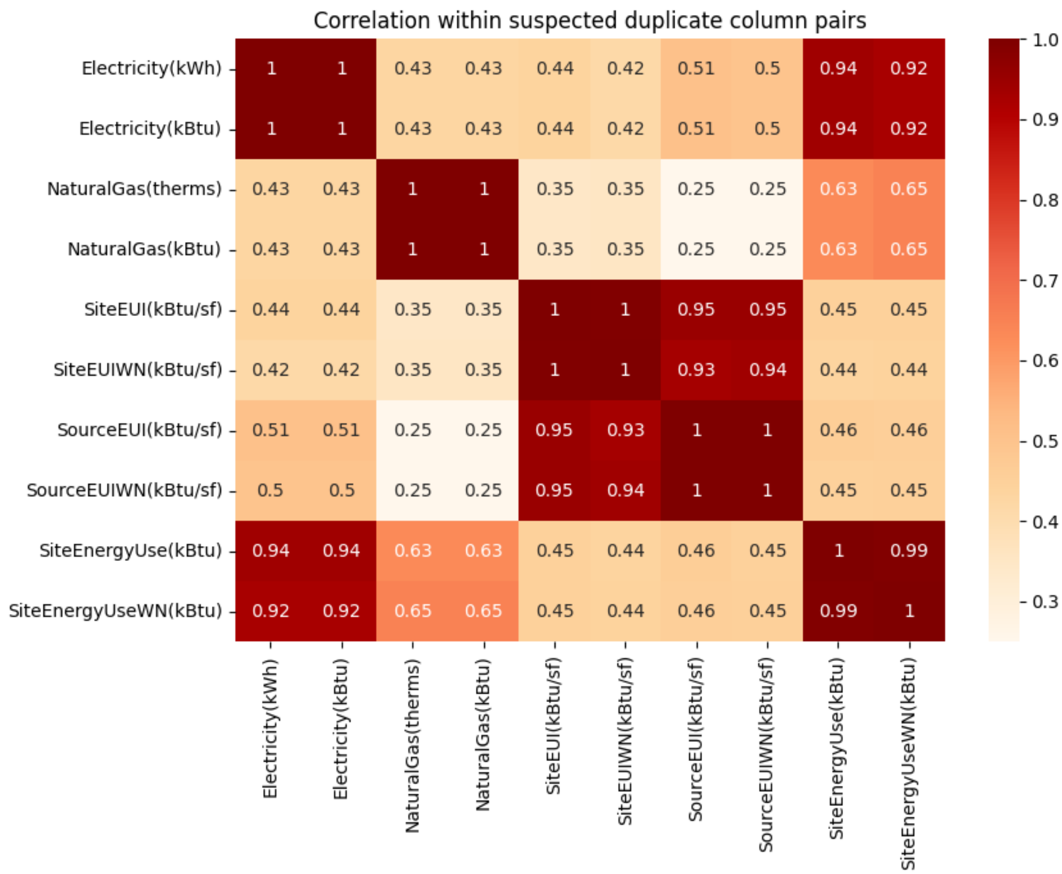
An “Outlier” column was discovered and used to remove all rows that had already been flagged as outliers by the city. The Location column was categorized as an object so this was further investigated and revealed that the latitude and longitude of the buildings were stored as one object. In the cleaning process, the latitude and longitude were separated and values were stored in separate columns as floats. Nine columns were missing over 50% so they were dropped immediately. The regression target was also identified to have 23% of values missing, so those specific rows were dropped.

The describe() summary surfaced impossible values (ie buildings that had Gross Floor Area > 0, but had negative or zero energy use). These rows were also removed.

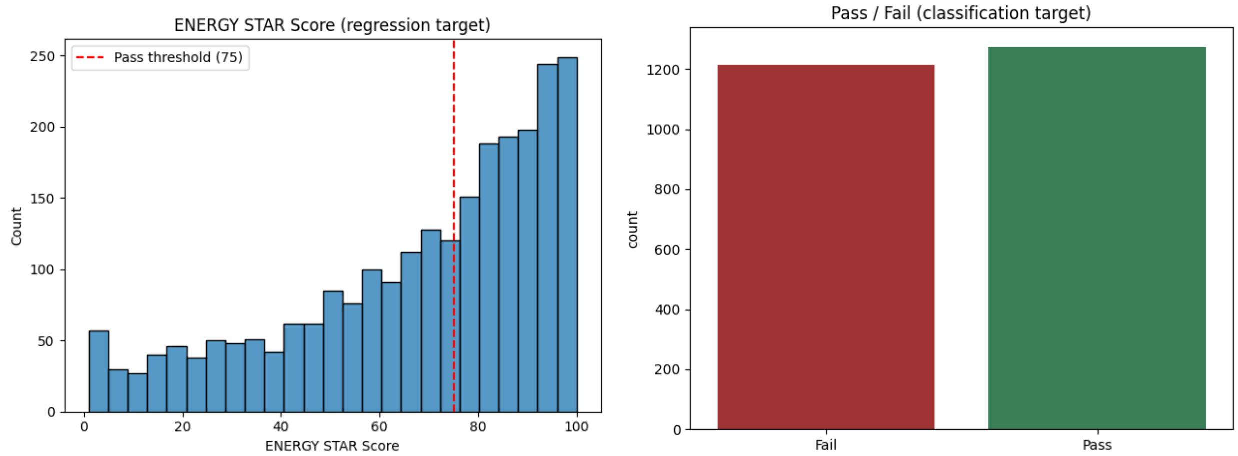
Finally, skew() was used to verify if any skew existed in the data and several columns showed a heavy right-skew. This was noted to be managed in the column transformation steps.

Visual Methods

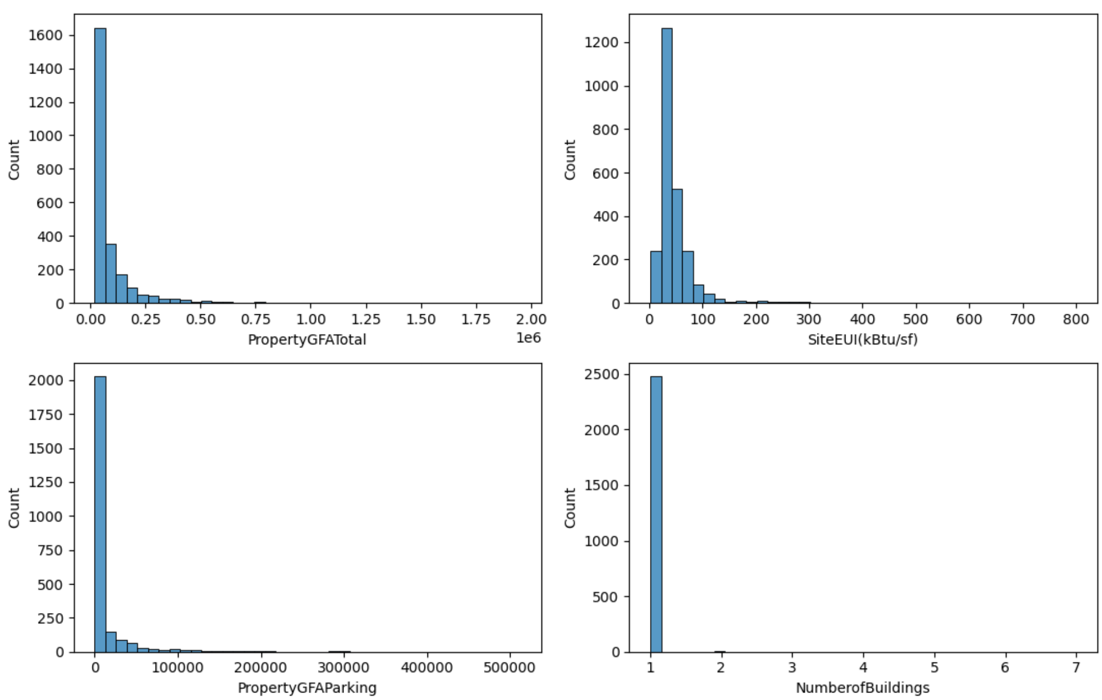
A heatmap was first used to validate suspected duplicate column pairs between energy metrics. The matrix confirmed duplication and those columns were dropped.



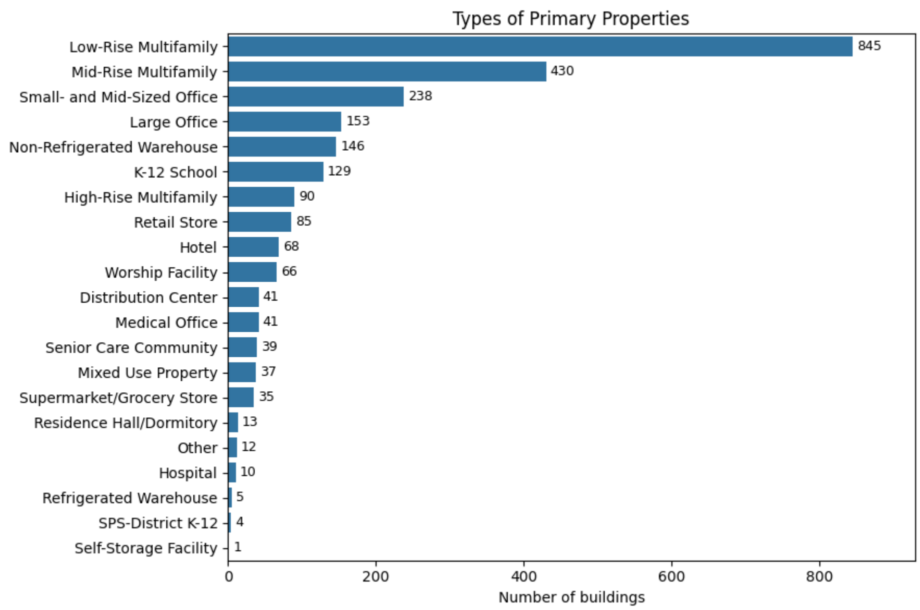
A histogram of the Energy Star score was used to visualize the distribution of the regression target to identify any skew in the target, and a bar chart of the derived Energy Star certification was used to highlight any imbalances in the target. The regression target displayed a left skew, which contributed to the decision of using a decision tree model (Gradient Boosting) in addition to the linear model. The classification target was almost perfectly balanced.



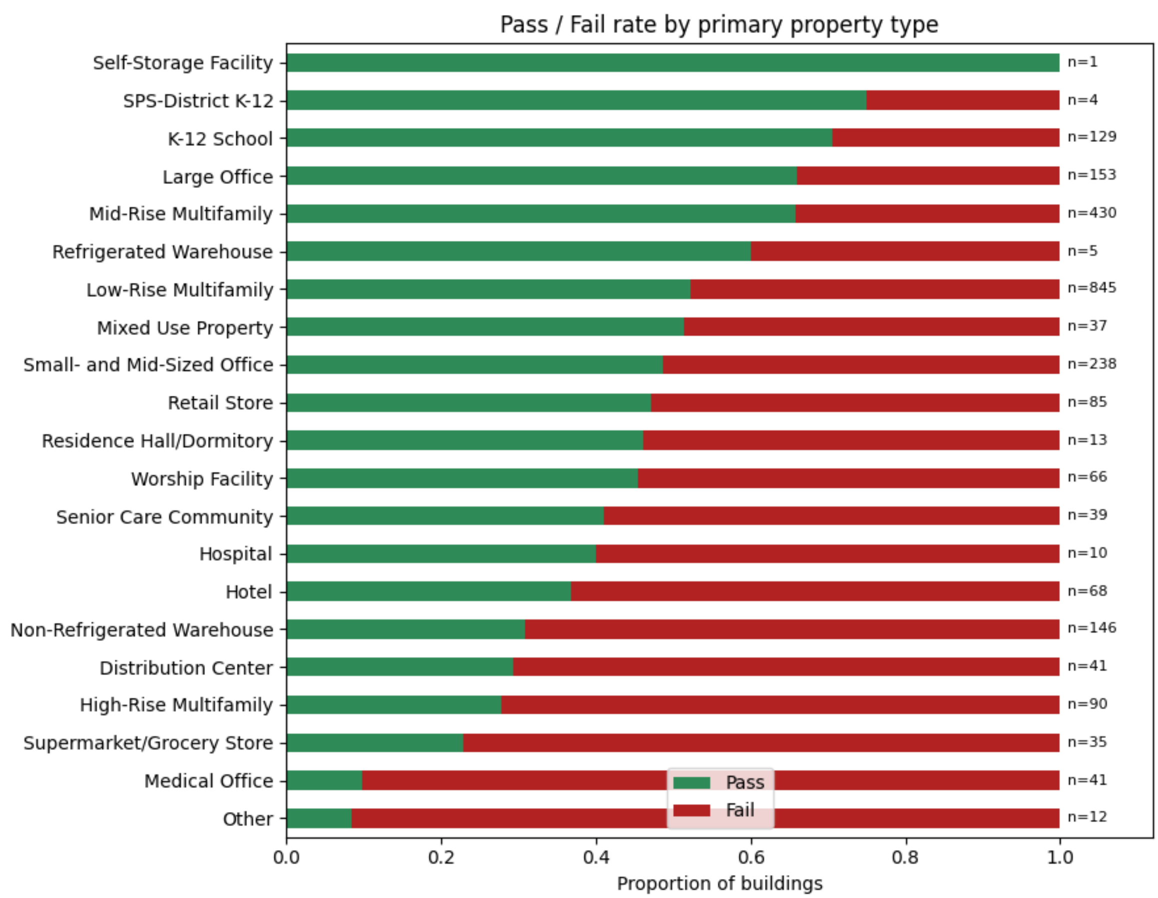
Histogram plots of a few features were also used to visualize the skew identified by the statistical skew() method.



A horizontal bar chart was used to display the possible categories in the Primary Property Type feature. This was to identify what type of encoding would be best on this feature. The large list of categories shown, along with the fact that a few of them had far smaller values than the others, led to the decision to group the “rare” categories into an “Other” option. This was managed in the column transformation preprocessor.



A 100% stacked bar chart of pass / fail by primary property type was also to identify any potential predictive signal.



Finally, another heatmap was used to identify correlation between numeric features. Several of the features displayed correlation with one another, leading to the decision to use Ridge to manage multicollinearity without losing interpretability. There was also correlation identified with the target variable and a couple features, which inspired the decision to test the models on a subset of the data that removed those correlated features.

Data Mining Tasks

Various data mining tasks were performed in this project, including data transformation, prediction, and classification. Both prediction and classification were demonstrated for the sake of applying both types of modelling learned in this course. They are **independent** supervised learning exercises, as the classification target is derived directly from the regression target.

Note that dimensionality reduction (ie PCA) was not performed as interpretability was a desired goal of this project.

Data Transformation

A custom transformer was developed to group the small primary property types into an "Other" category to reduce noise and variance.

A SimpleImputer using median imputation was used to fill in the missing values that weren't dropped. Median was selected over Mean due to the heavy skew in the features.

A PowerTransformer (yeo-johnson) was used to correct the heavy right skew while also scaling the data in one step.

OneHotEncoder was used to encode the remaining categorical features.

All transformers were implemented inside a scikit-learn Pipeline and ColumnTransformer fit on the training split only, so that imputation statistics, encoder categories, scaling parameters, and the rare-category list never leak information from the test set. This pipeline was then used across all modelling exercises.

Aside from the removal of city-flagged outliers, no other trimming was applied as the PowerTransformer could manage skewed data and the decision tree models selected for the project are largely invariant to feature outliers.

Regression Models

As one of the goals of this project was to understand which factors drive the Energy Star score, interpretable models were selected with readable coefficients / importances. Ridge was chosen over plain linear regression because of the multicollinearity between energy features as seen in the EDA. Furthermore, Ridge was selected over Lasso because Lasso tends to arbitrarily retain one feature from a correlated group and zero out the others, impacting interpretability. Ridge's L2 penalty on the other hand, shrinks correlated coefficients together and stabilises them while keeping every feature in the model.

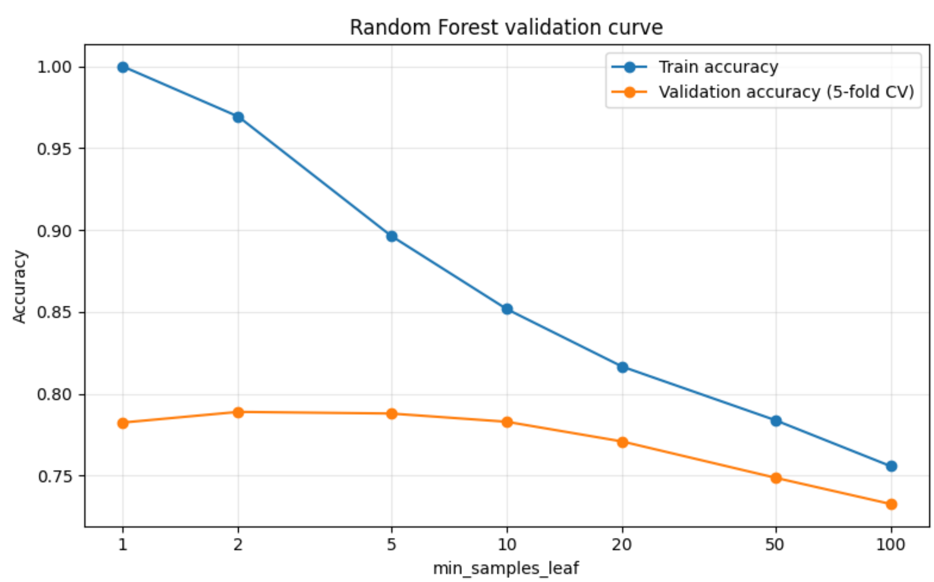
Cross validation (RidgeCV) over a log-space grid (selected to test a broader range of parameters more efficiently) was used to determine the regularization strength. Then, 5-fold cross validation was applied to evaluate model performance before the final scoring.

As a comparison to Ridge, a Gradient Boosting Regressor was applied due to the suspicion of non-linear relationships that Ridge could not capture, as well as the skew seen on the target variable. The regressor was fit on the same pipeline and train/test split as Ridge for a direct comparison.

Classification Models

Similar to the thought process for regression, two models were compared: an interpretable linear baseline (Logistic Regression), followed by Random Forest to test whether non-linearity and feature interactions would improve the results, while still providing feature importances. Both were wrapped in the same preprocessing pipeline and fit on the same train/test split as the regression models.

To prevent overfitting, Random Forest complexity was tuned with a validation curve over min_samples_leaf (using 5-fold cross-validation, not the test set to prevent train/test contamination). A min_samples_leaf value of 10 was selected as it minimized the training to validation gap, without penalizing the validation accuracy.

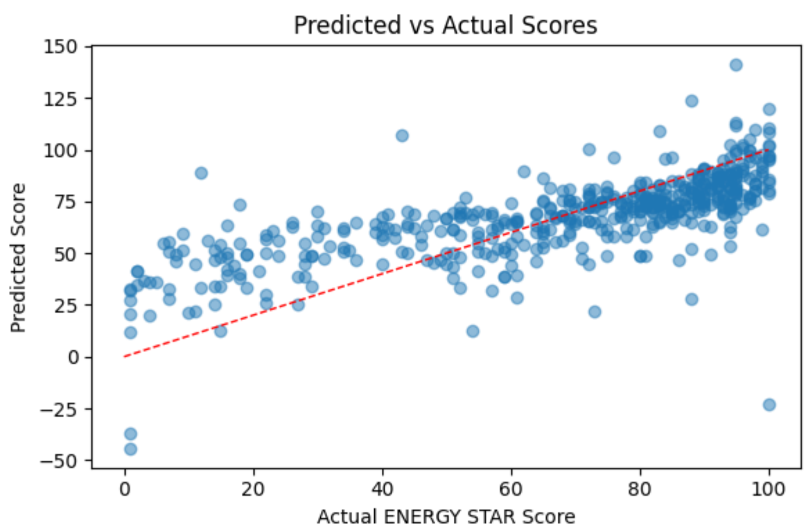


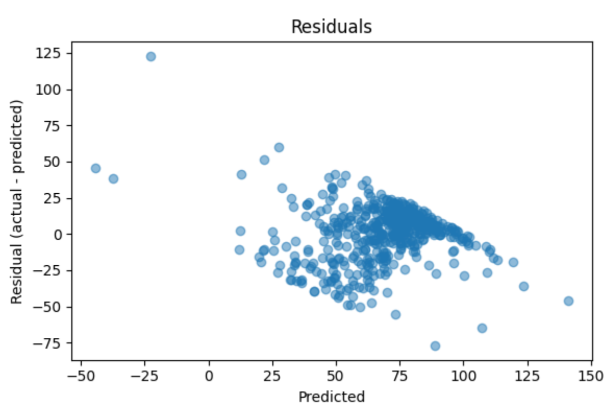
To address the target leakage caused by the significant correlation between Source / Site EUI and the target variable, the classification models were also applied on a subset of the dataset that included solely physical building characteristics. Removing the energy features introduces more of a non-circular question, and the gap in performance between the full and building-only models quantifies how much predictive power came from this target leakage.

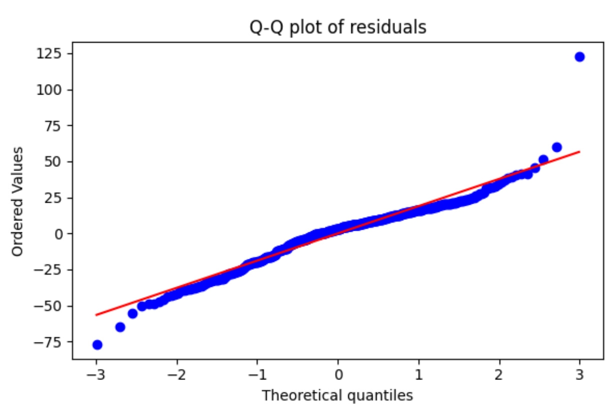
Performance Evaluation

Regression

Primarily, a train/test R^2 comparison was used to validate the Ridge model was not overfitting. Diagnostic plots (residuals, QQ plot) were used to verify homoskedasticity and normality of the Ridge model.







The QQ plot shows a relatively normal distribution. The ends do show a few outliers, confirming that most of the errors are small, but there are a few large ones causing less than optimal results. We can also see heteroskedasticity from the residuals vs predicted plot, showing that there are non-linear relationships being missed by the model. This is also reflected in the predicted vs actuals plot.

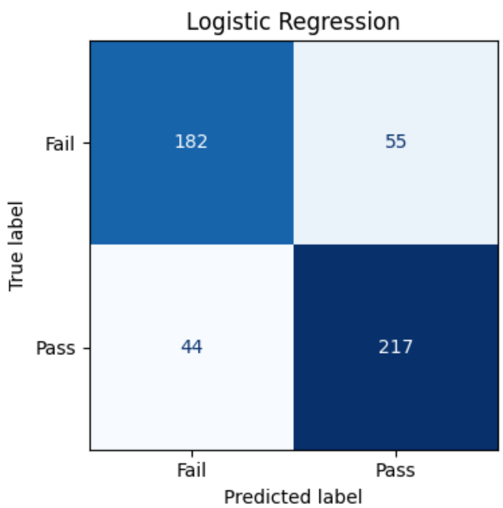
Performance of both Ridge and Gradient Boosting were then compared using the metrics R^2, RMSE, and MAE. The difference in RMSE and MAE confirmed the conclusion made by the QQ plot. Also to note, the cross-validation estimate is the lowest, which is expected: each CV fold trains on only four-fifths of the training data (about 64% of the full dataset), whereas the final model is refit on the entire training set, so k-fold CV is mildly pessimistic. Comparing the two models, Gradient Boosting raised the test R^2 from **0.50 to 0.63** and cut error from RMSE ~19.3 to ~16.6 score points. This confirms that there is meaningful non-linearity and feature interactions that Ridge could not capture.

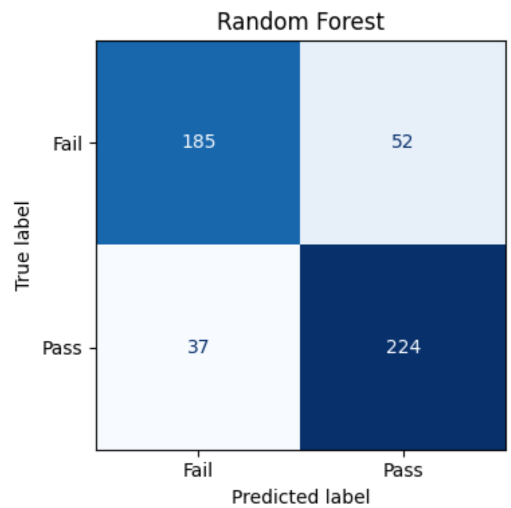
	Ridge	Gradient Boosting
Train R^2	0.506	0.689
CV R^2	0.466	0.543
Test R^2	0.504	0.632
Test RMSE	19.310	16.619
Test MAE	14.468	12.034

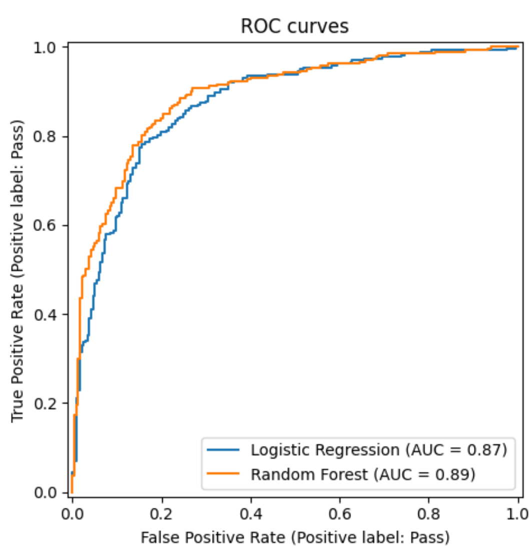
While Gradient Boosting performed better than Ridge, an R^2 of 0.63 is still not ideal, so non-linearity was not the only inhibiting factor. From research, it seems there are key operational variables missing in this dataset that would improve the prediction performance (ie operating hours, occupancy, etc). The score is also calculated on a comparison basis by primary property type, so while Gradient Boosting captures these feature interactions, it could be further improved by adding a feature that explains relative_EUI within its property type. This would give it the added information of a property type's mean EUI and standard deviation, giving a clearer picture of peer comparison.

Classification

Classification was evaluated with accuracy, precision, recall, F1, and ROC-AUC, and performance was further visualized with confusion matrices and ROC curves. Similar to the regression models, Random Forest consistently performed better than Logistic Regression due to the ability to capture non-linear relationships and feature interactions. In contrast to the regression outcomes, however, there was not a large gap in performance. This is partly due to the fact that the target for classification has more room for error (a simple pass / fail compared to an exact score prediction), as well as the consideration that Gradient Boosting is typically a stronger model since it iteratively improves the model compared to Random Forest which simply takes an average.







As previously mentioned, the models were also applied on a building-only subset of the dataset to address target leakage and identify any other potentially interesting insights related to building characteristics. The same metrics were used to evaluate the models, and as expected – due to the significant correlation with energy features – performance dropped. This drop in accuracy reflects the share of predictive power that came from *how the building uses energy* rather than *how it's built*. The weak signal that remains is what a building owner can anticipate from physical attributes alone.

	Logistic Regression (Full)	**Logistic Regression (Building)**	**Random Forest (Full)**	**Random Forest (Building)**
Accuracy	0.801	0.614	0.821	0.604
Precision	0.798	0.626	0.812	0.631
Recall	0.831	0.655	0.858	0.590
F1	0.814	0.640	0.834	0.610
ROC-AUC	0.874	0.648	0.891	0.688

One thing to note: on the building-only model, Logistic Regression actually performed slightly better than Random Forest on *some* metrics (accuracy, recall, and F1). This is because these metrics are all computed at a fixed prediction cutoff of 0.5, whereas ROC-AUC is threshold-free as it measures how well a model performs across all probability cutoffs. When signal is weaker, Random Forest probabilities get compressed toward the middle, so the 0.5 cutoff is less ideal.

Project Results

Feature Drivers

Both Ridge Regression and Logistic Regression coefficients revealed the peer comparison calculation that was mentioned earlier. In the full model, the Supermarket/Grocery Store primary property type coefficient was strongly positive, but that was conditional on energy use: if Source EUI is held constant between all property types, a supermarket would have higher odds of passing than a non-refrigerated warehouse or a school for example. This means the Energy Star score is property-type normalized (supermarkets are expected to have higher energy use so other property types would be more strongly penalized at the same level). With the energy features removed, that confounder is gone, so the building-only coefficients move back toward the raw pass-rate ordering of property types (e.g., schools typically have higher odds of passing than supermarkets).

Predictor	coefficient
cat__PrimaryPropertyType_Supermarket/Grocery S...	4.5059
cat__PrimaryPropertyType_Non-Refrigerated Ware...	-2.9862
cat__PrimaryPropertyType_Worship Facility	-2.3595
cat__PrimaryPropertyType_Large Office	2.3402
cat__PrimaryPropertyType_Small- and Mid-Sized ...	1.7569
cat__PrimaryPropertyType_Retail Store	1.6394
num__SourceEUI(kBtu/sf)	-1.4422
num__GHGEmissions(MetricTonsCO2e)	1.2653
num__SiteEUI(kBtu/sf)	-1.1045
cat__PrimaryPropertyType_Senior Care Community	1.0525
cat__BuildingType_Multifamily HR (10+)	-1.0471
cat__BuildingType_SPS-District K-12	1.0210
cat__PrimaryPropertyType_Mixed Use Property	0.9509
cat__Neighborhood_GREATER DUWAMISH	-0.8737
num__PropertyGFATotal	0.7858

Full Dataset

Predictor	coefficient
cat__BuildingType_SPS-District K-12	1.2482
cat__PrimaryPropertyType_Medical Office	-1.2064
cat__PrimaryPropertyType_Large Office	0.9230
cat__BuildingType_Multifamily HR (10+)	-0.8731
cat__Neighborhood_CENTRAL	-0.6664
cat__BuildingType_Nonresidential COS	-0.6493
cat__PrimaryPropertyType_K-12 School	0.6491
cat__Neighborhood_MAGNOLIA / QUEEN ANNE	0.6301
cat__Neighborhood_SOUTHEAST	-0.6016
cat__BuildingType_Multifamily LR (1-4)	0.5530
cat__BuildingType_Multifamily MR (5-9)	0.5468
cat__Neighborhood_LAKE UNION	0.5031
cat__PrimaryPropertyType_Supermarket/Grocery S...	-0.4855
cat__PrimaryPropertyType_Small- and Mid-Sized ...	0.4677
cat__Neighborhood_SOUTHWEST	0.4113

Building Dataset

Using the full dataset, the Random Forest feature importances identified SourceEUI and SiteEUI as the dominant drivers, as expected. Removing the energy columns redistributed that importance onto the physical features: building age became the single most important predictor, followed by a tight cluster of size measures (LargestPropertyUseTypeGFA, PropertyGFABuilding(s), PropertyGFATotal).

Predictor	importance
num__SourceEUI(kBtu/sf)	0.2373
num__SiteEUI(kBtu/sf)	0.1643
num__Electricity(kBtu)	0.0658
num__GHGEmissionsIntensity(kgCO2e/ft2)	0.0658
num__SiteEnergyUse(kBtu)	0.0427
num__PropertyGFATotal	0.0403
num__building_age	0.0399
num__GHGEmissions(MetricTonsCO2e)	0.0361
num__LargestPropertyUseTypeGFA	0.0355
num__PropertyGFABuilding(s)	0.0331
num__PropertyGFAParking	0.0301
num__NaturalGas(kBtu)	0.0281
num__NumberofFloors	0.0229
cat__BuildingType_NonResidential	0.0173
cat__PrimaryPropertyType_Low-Rise Multifamily	0.0158

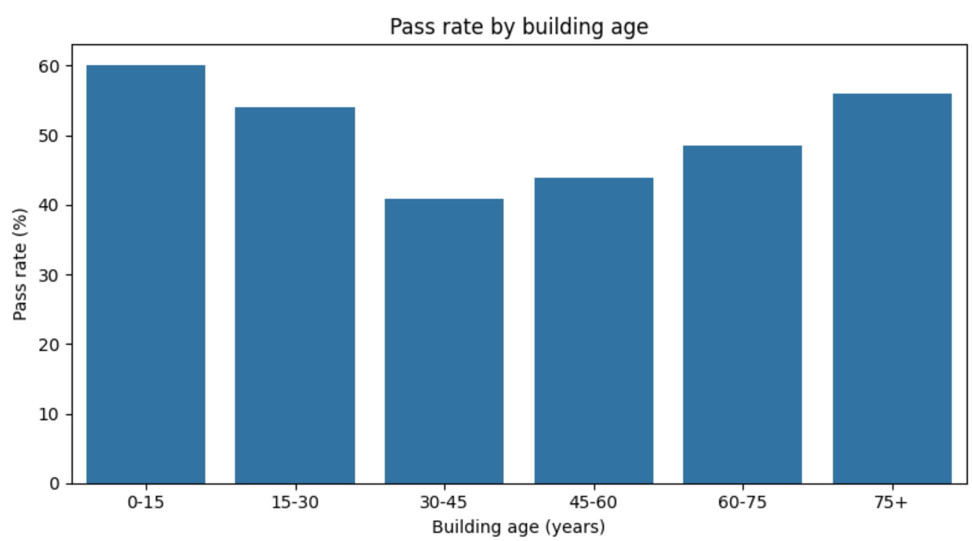
Full Dataset

Predictor	importance
num__building_age	0.1310
num__PropertyGFABuilding(s)	0.1117
num__PropertyGFATotal	0.1115
num__LargestPropertyUseTypeGFA	0.1053
num__PropertyGFAParking	0.0728
num__NumberofFloors	0.0596
cat__Neighborhood_MAGNOLIA / QUEEN ANNE	0.0411
cat__BuildingType_Multifamily MR (5-9)	0.0391
cat__BuildingType_SPS-District K-12	0.0383
cat__PrimaryPropertyType_Mid-Rise Multifamily	0.0364
cat__PrimaryPropertyType_K-12 School	0.0258
cat__BuildingType_NonResidential	0.0238
cat__PrimaryPropertyType_Large Office	0.0191
cat__Neighborhood_GREATER DUWAMISH	0.0180
cat__BuildingType_Multifamily HR (10+)	0.0179

Building Dataset

Interestingly, age had a near-zero coefficient in the logistic model, highlighting that there is a non-linear relationship (U-shaped) with the pass rate that logistic regression cannot capture with a single linear age term - newer and older buildings have higher odds of passing than buildings built between 1940-1985. This is visualized in the chart below.

This trend is explained by general industrialization concepts and can also be seen in country's per capita emissions. Emissions are low pre-industrialization due to the lack of access to cheap fossil fuels and electricity, levels rise with modernization, then reduce again with newer climate-conscious technologies and policies.



Performance Summary

To summarize, Random Forest classification on the full dataset performed the best. This is because it was able to capture non-linear relationships and feature interactions, it had more room for error with the classification target, and the score is heavily derived from energy use so keeping the full dataset produced the best outcomes.

Performance overall was limited by missing operational data like occupancy, operating hours, etc. Incorporating this information into the dataset would likely improve the performance of the models and produce more interesting feature importance results.

Results Summary

The Energy Star score is heavily dependent on energy consumption. However, it is not a completely linear relationship, as the score is normalized by property type (energy consumption is measured against the energy consumption of peer property types). There is also a small correlation with building age that impacts energy efficiency results.

Impact of the Project Outcomes

Insights

The models turn a raw disclosure dataset into actionable understanding. Because the analysis shows that energy-use intensity is both the dominant driver of the score and the factor owners can actually change, it points clearly to where improvement effort pays off. The main message to building owners would be to focus efforts on operational efficiency improvements (e.g., HVAC, lighting, etc) to improve their Energy Star scores. Since scores are calculated based on performance within the primary property type, it would also be beneficial to understand energy use from peer buildings to potentially improve their score.

Additionally, the model performance and further research on Energy Star scores revealed missing information (operational variables) that would yield better results if included in the dataset.

Future Work

A concrete path to improving the regression would be to add a feature that would more directly explain the peer-group structure the EPA uses (energy intensity feature expressed relative to each property type's mean - a within-type z-score). This would let even a linear model approximate the property type normalization that defines the score, capturing structure the tree models currently learn only implicitly.

Further tuning the gradient-boosting regressor could also help improve performance.

Other Considerations

While this exercise was interesting in that it revealed helpful insights about property type normalization and non-linearity with building age, overall it was rather redundant due to the target leakage and lack of variables that would have added a more interesting layer to the analysis.

After further research on ESPM's Energy Star score calculation, I discovered that the score is actually derived from an **Efficiency Ratio**. This ratio is calculated by dividing Actual Source EUI by Predicted Source EUI, where the Predicted Source EUI is produced by the EPA's regression models. This ratio is then mapped to a lookup table that scores buildings against their national peers. With access to the appropriate data, a less redundant predictive modelling exercise could be to predict ESPM's Predicted Source EUI to allow building owners to determine their performance in advance and adjust as required, before being officially evaluated by the EPA.